

Rohan Tan Bhowmik

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Summary:

Senior at Stanford University pursuing Electrical Engineering and Computer Science with a focus on artificial intelligence. Researcher at the Intelligence through Robotic Interaction at Scale (IRIS) Lab under Professor Chelsea Finn. Teaching humanoid robots to play sports and compete against humans. Co-president of Stanford Student Robotics and software lead for Robosub, Stanford's autonomous underwater vehicle project. Hands-on experience in applying AI to robotics, health, and human-computer interaction. Experienced with machine learning compilers for optimizing foundation model performance on heterogenous computing architectures. Deeply interested in brain-inspired computational architectures for embodied intelligence.

Education:

Stanford University

B.S., Electrical Engineering, Class of 2027 (GPA: 3.92/4.00)

M.S., Computer Science, Expected Graduation Dec 2027

Experience:

Researcher, Stanford IRIS Lab

- Human-out-of-the-loop Robot Learning (Jun 2026 – present)
 - Developing a task-and-scene proposal pipeline for autonomous self-play, enabling robot policies to acquire diverse manipulation skills with minimal human supervision.
 - Designing a framework for continuous skill acquisition (day-time autonomous exploration and reset, night-time trajectory-based training) across a multi-arm setup.
- Correction and Reinforcement from Evidence (May 2026 – present)
 - Developed data collection strategy prioritizing failure-distribution coverage and counterfactual demonstrations to optimize policy training efficiency.
 - Designed dual-signal training loop combining human corrections and policy rollouts to jointly update a policy actor and Implicit Q-Learning (IQL) critic.
- Learning in Partially Observable Environments (Oct 2025 – present)
 - Investigated policy training under partial observability, addressing self-occlusion failure modes (e.g., arm blocking camera line-of-sight during manipulation).

Independent Research

- Humanoid Robots (Jan 2026 – present)
 - Training a Unitree G1 humanoid to play table tennis at a competitive/near-professional level via human demonstration and reinforcement learning with sim-to-real transfer.
 - Teaching humanoid robots sport and movement skills, including football throwing, soccer kicking, and dancing.

- Spiking Neural Networks (SNNs) (Apr 2026 – present)
 - Extended Hodgkin-Huxley neuron dynamics to model biological neurogenesis, enabling a spiking neural network to grow and prune neurons over time as it learns.
 - Cut catastrophic forgetting by ~3x versus a fixed-size network by protecting mature neurons during continual learning, as validated via an ablation study.
- Quantum Optical Convolutional Neural Networks (QOCNNs) (Jun 2019 – Mar 2020)
 - Created a novel framework to simulate the training of convolutional neural networks on quantum computing systems.
 - Demonstrated >100x boosts in computational and memory efficiency for training a novel Quantum Optical Convolutional Neural Network, as compared with a classical Convolutional Neural Network.

Co-President, Stanford Student Robotics Club (Sep 2025 – Jun 2026)

- Led student engagement with robotics on campus by organizing project showcases, company events, and hackathons, growing participation and industry partnerships across the club.

Software Lead, Stanford Autonomous Underwater Vehicle Project (Oct 2024 – Jun 2026)

- Led the design and implementation of a ROS 2 based autonomy stack for a new autonomous underwater vehicle, integrating perception, localization, and path-planning.
- Managed a cross-disciplinary software team of ~15 engineers, integrating mechanical, electrical, and research requirements into a unified, field-validated robotics platform for competitions and scientific missions.

Software Engineering Intern, AMD AI Group (Jun 2024 – Sep 2025)

- Specialized in machine learning compilers, optimizing model performance across diverse hardware architectures and reducing latency in large-scale deployments.
- Contributed to Wave, a GPU domain-specific language, by designing and extending custom kernels to improve compilation efficiency and boost inference performance on servers.

Research Intern, Stanford University

- California Wildfire Prediction System (Jun 2022 – Jan 2024)
 - Developed a wildfire prediction system based on a novel machine-learning framework, successfully predicting 85% of major wildfires in California up to 2 weeks in advance.
 - Compiled a comprehensive wildfire dataset with >37 million data points including environmental, meteorological, and geological factors.
- AI-assisted Facial Surgery Planning and Evaluation (Jun 2021 – May 2022)
 - Developed AI algorithms to aid surgical planning for correcting facial structural deformities that cause debilitating breathing, sleeping, and sensory disorders in patients.
 - Achieved >95% accuracy in classifying patient conditions with machine learning training.
- COPD Exacerbation Prediction (Jun 2020 – May 2021)
 - Developed a multi-modal solution for predicting the climate's impact on exacerbation risks of respiratory diseases, such as COPD.
 - Created a new AI architecture that combines key attributes of both convolutional and recurrent neural networks with 93% cough detection accuracy.
 - Proposed a predictive model based on retrospective medical studies to assess respiratory exacerbation risk.

Publications:

Peer-Reviewed Journal Papers:

1. T. H. Nguyen, C. Akdis, A. Baccarelli, S. Banerjee, **R. T. Bhowmik**, et al, "Leveraging Artificial Intelligence in Allergy, Asthma, and Immunology With Environmental Exposures," *Allergy*, 2026, pp. 1–16. ([link](#))
2. C. Ding, **R. T. Bhowmik**, "Artificial Intelligence in Multimedia Content Generation: A Review of Audio and Video Synthesis Technique," *Journal of the Society for Information Display*, 34, 2026, pp. 49–67, Invited Paper. ([link](#))
3. **R. T. Bhowmik**, Y. S. Jung, J. Aguilera, M. Prunicki, K. Nadeau, "A Multi-Modal Wildfire Prediction and Personalized Early-Warning System Based on a Novel Machine Learning Framework," *Journal of Environmental Management*, **341**, 117908, 2023. ([link](#))
4. **R. T. Bhowmik**, C. Kandathil, S. P. Most, "Automating the Standardized Cosmesis and Health Nasal Outcomes Survey (SCHNOS) Classification with Convolutional Neural Networks," *Facial Plastic Surgery & Aesthetic Medicine*, PMID: 36749153, 2023. ([link](#))
5. **R. T. Bhowmik**, S. P. Most, "A Personalized Respiratory Disease Exacerbation Prediction Technique Based on a Novel Spatio-Temporal Machine Learning Architecture and Local Environmental Sensor Networks," *Electronics*, **11** (16), 2562, 2022. ([link](#))
6. R. Parthasarathy, **R. T. Bhowmik**, "Quantum Optical Convolutional Neural Network: A Novel Image Recognition Framework for Quantum Computing," *IEEE Access*, **9**, 103337, 2021. ([link](#))
7. **R. T. Bhowmik**, Y. S. Jung, J. Aguilera, M. Prunicki, K. Nadeau, "California Wildfire Forecasting Inventory - Extensive Wildfire Prediction Dataset for Artificial Intelligence," *Fire* (under review).

Preprints:

8. **R. T. Bhowmik**, Y. S. Jung, J. Aguilera, M. Prunicki, K. Nadeau, "California Wildfire Forecasting Inventory - Extensive Wildfire Prediction Dataset for Artificial Intelligence," arXiv:2509.11015, 2025.
9. **R. T. Bhowmik**, "A Multi-Modal Wildfire Prediction and Personalized Early-Warning System Based on a Novel Machine Learning Framework," arXiv:2208.09079, 2022.
10. **R. T. Bhowmik**, "A Multi-Modal Respiratory Disease Exacerbation Prediction Technique Based on a Spatio-Temporal Machine Learning Architecture," arXiv:2103.03086, 2021.
11. R. Parthasarathy, **R. T. Bhowmik**, "Quantum Optical Convolutional Neural Network: A Novel Image Recognition Framework for Quantum Computing," arXiv:2012.10812, 2020.

Conferences:

12. **R. T. Bhowmik***, S. Sun*, E. McElhinney, V. Alexopoulos, S. Hickmann, F. Hoffmann, K. Song, R. Sato, L. Skaling, A. Krinos, C. Ogaki, O. Khatib, "3D Reconstruction of Underwater Features in Lake Tahoe with an Autonomous Underwater Vehicle," International Conference of Computer Vision, 2025. ([link](#))
13. **R. T. Bhowmik**, "A Multi-Modal Wildfire Prediction and Personalized Early-Warning System Based on a Novel Machine Learning Framework," Stanford BigEarth Hackathon, 2022.